SOFTWARE REQUIREMENTS & SYSTEM SPECIFICATION

Pre-Order Food Delivery Application

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Appendix A1 – A6 carries the full specification tables: actors and business goals, the data model, all 30 functional requirements with their acceptance criteria, all 16 business rules and all 36 unresolved items.

01

Document Purpose & Scope

§1 — §2

DOCUMENT OVERVIEW §1

Purpose and derivation contract

This document converts the validated Requirements Elicitation Document into a software-oriented specification suitable for system design, AI-assisted implementation, API and UI development, database design, testing, security review, requirements traceability and stakeholder validation.

No new business requirements.

The specification introduces none. Everything below is derived from the elicitation document, and each item carries the identifier it came from.

Implementation detail is marked, not assumed.

Where detail is necessary but the elicitation document does not define it, the specification labels it Proposed or Unresolved rather than inventing a decision.

HOW TO READ THE SPECIFICATION §1.1

VALIDATED Carried unchanged from the elicitation document. Authoritative.

PROPOSED A design decision added by the specification for implementability. Not a validated business requirement.

UNRESOLVED A decision the requirements do not make. Must not be invented by an implementation agent.

INTENDED AUDIENCE §1.3 Software developers · AI coding agents · Software architects · QA and test engineers · Security engineers · UI/UX designers · Restaurant stakeholders · Platform administrators · Project supervisors · Requirements engineers

Scope of the initial version

IN SCOPE — 20 CAPABILITIES §2.1

01 Restaurant-provisioned customer access

02 Restaurant-specific customer credentials

03 Restaurant staff access

04 Delivery-person access

05 Restaurant management

06 Menu management

07 Menu-item availability

08 Morning pre-ordering

09 Order calculation and submission

10 Customer and organization verification

11 Restaurant order preparation

12 Restaurant delivery-person assignment

13 Delivery status management

14 Payment on delivery

15 Tunisian electronic payment integration

16 Order and payment status recording

17 Customer order visibility

18 Notifications where services are integrated

19 Platform administration

20 Security, privacy, logging, testing, monitoring

OUT OF SCOPE §2.2

— AI demand prediction, recommendations and decision making

— Platform-employed delivery personnel

— Platform-provided delivery equipment

— Assumed support for every Tunisian payment provider

— Automatic resolution of rejected, cancelled or failed deliveries

— Refund logic and cancellation logic

— Pickup workflow, unless stakeholders approve it

> Everything on this list is excluded from the initial version. AI is excluded by assumption A-13; delivery ownership is excluded by A-04 to A-07 and BR-06, BR-07.

Definitions and normative terms

Customer
Person who orders food through the platform.

Restaurant
Food provider using the platform.

Restaurant Staff
Personnel managing restaurant operations and orders.

Restaurant Manager
Person responsible for restaurant-level management.

Delivery Person
Restaurant-provided person responsible for physical delivery.

Platform Administrator
Person managing the platform globally.

Pre-Order
Food order submitted before the restaurant's defined ordering deadline.

Delivery Equipment
Equipment supplied by the restaurant for food delivery.

Target Organization
Workplace, campus or institution associated with a customer.

E-Payment
Supported third-party electronic payment service.

Ordering Deadline
Restaurant-defined time after which an order cannot be submitted.

SOURCE DOCUMENT §1.5

The normative baseline

30	Functional requirements	FR-01 — FR-30
24	User needs	UN-01 — UN-24
9	Business goals	BG-01 — BG-09
16	Business rules	BR-01 — BR-16
15	Assumptions	A-01 — A-15
37	Open questions	Q-01 — Q-37
11	Stakeholders	Parties affected by the system

> The sole normative source is the validated Requirements Elicitation Document. Every requirement in this specification is one of these items restated for software construction — none is new, and none is dropped. AI is explicitly out of scope for the initial version.

02

Actors & System Context

§2 · §4 – §5

Actors and roles

ACTOR	RESPONSIBILITIES
Customer	Browse restaurants, view menus, place orders, pay, receive food, view status.
Student	Customer who studies at a target institution.
Employee / worker	Customer who works at a target organization.
Restaurant owner	Owns the restaurant business.
Restaurant manager	Manages restaurant operations.
Restaurant staff	Manages menus and orders, and prepares food.
Delivery person	Performs the restaurant's delivery.
Platform administrator	Manages platform-level users, restaurants and operational issues.
Platform owner	Owns the platform and the business.
Payment provider	Processes supported electronic payments.
Workplace / campus organization	The organization associated with a customer.

The platform coordinates. The restaurant delivers.

Restaurant responsibilities

Prepare the food. Provide the delivery personnel. Provide the delivery equipment. Assign its delivery personnel. Perform the physical delivery.

Platform responsibilities

Coordinate information. Receive and manage orders. Provide the restaurant, customer and delivery interfaces. Coordinate delivery information. Record statuses. Integrate supported payment services.

> The platform does not perform physical delivery, and it employs nobody to perform it. Every physical act in the chain belongs to the restaurant.

BOUNDARY §2.3

Inside and outside the system

INSIDE THE SYSTEM — 12

- Customer accounts and credentials
- Restaurant relationships
- Restaurant accounts
- Staff access
- Delivery-person access
- Menus
- Orders
- Order status
- Delivery assignment
- Payment status
- Customer organization information
- Administrative functions

OUTSIDE THE SYSTEM — 7

- Physical food preparation
- Physical transportation
- Delivery personnel employment
- Delivery equipment provision
- Third-party payment processing
- SMS and OTP infrastructure
- External notification infrastructure

The system records and coordinates. Preparation, transport, employment, equipment, payment processing and notification infrastructure all stay outside it.

External systems and integration status

- Tunisian electronic payment providers
- SMS and OTP services
- Email services
- Push-notification services
- External support systems

> The document specifies these as services the platform may integrate. Integration with any particular provider remains dependent on stakeholder decisions, so none of them is a fixed interface.

System context: capabilities and interfaces

REQUIRED CAPABILITIES §5.4

- Customer access
- Restaurant management
- Staff access
- Delivery operations
- Orders
- Payments
- Administration
- External payment providers
- Notification services where integrated

The validated requirements establish API capabilities but do not define REST paths, schemas or rate limits. The exact production API technology is Proposed, not a validated requirement.

PER-ACTOR INTERFACES §13

Customer

Access and login, restaurant list, menu, cart and order, payment selection, confirmation, order status.

Restaurant

Dashboard, menu management, incoming orders, customer verification, delivery personnel, assignment, order and payment status.

Delivery person

Assigned orders, order details, delivery information, delivery status, delivery confirmation.

Administrator

Users, restaurants, staff, operational monitoring, system events.

Screen counts and exact UI copy are Proposed. The administrator surface beyond FR-29 and FR-30 remains dependent on stakeholder validation.

03

Vision & Requirement Map

§3 · §6

GOALS §3, §3.1

Vision and business goals

The product gives customers in high-density areas a structured way to pre-order food during the morning and receive it during lunch, reducing the need for lunch-time ordering and helping restaurants plan preparation and delivery.

BG-01
Reduce lunch-hour waiting time.
Ordered before the peak.

BG-02
Move ordering before lunch.
Pressure leaves the lunch hour.

BG-03
Help restaurants predict demand.
Met by order visibility, not by a model.

BG-04
Improve meal preparation planning.
Advance orders allow planning.

BG-05
Organize food delivery.
Assignments are structured.

BG-06
Support convenient payment methods.
Flexibility at the point of order.

BG-07
Restrict access by restaurant rule.
The restaurant decides who it serves.

BG-08
Improve order accuracy.
Digital orders remove miscommunication.

BG-09
Provide order visibility.
Customers can see progress.

No AI model is required to achieve BG-03 in the initial version. Order visibility alone satisfies it.

REQUIREMENT MAP §6

Six groups, 30 functional requirements

06 — FR-01 – FR-04, FR-29 – FR-30

Accounts and access

04 — FR-05 – FR-08

Restaurant & menu

07 — FR-09 – FR-15

Ordering

02 — FR-16 – FR-17

Verification

07 — FR-18 – FR-24

Delivery

04 — FR-25 – FR-28

Payment

FR-01 to FR-28 keep the numbering they carry in the elicitation document. FR-29 and FR-30 are the two provisioning requirements the specification adds, appended so that no existing identifier moves. Acceptance criteria for all 30 are in §17 and in appendix A3 and A4.

04

Functional Specification

§6

Accounts, access and menus

FR-01	The system shall support customer account creation.	§6.1 · BR-15, BR-16
FR-02	The system shall collect the customer's full name.	§6.1 · BR-05, BR-15
FR-03	The system shall collect the customer's phone number.	§6.1
FR-04	The system shall collect the customer's workplace or educational institution.	§6.1 · BR-04, BR-05, BR-15
FR-05	The system shall display available restaurants to the customer.	§6.2 · BR-04, BR-16
FR-06	The system shall allow customers to view restaurant menus.	§6.2 · BR-03
FR-07	The system shall allow restaurants to manage their menus.	§6.2 · BR-03, BR-09
FR-08	The system shall allow restaurants to mark menu items available or unavailable.	§6.2 · BR-02, BR-03
FR-29	The system shall allow the platform to provision accounts for restaurant staff.	§6.7
FR-30	The system shall allow restaurants to provision accounts for their delivery personnel.	§6.7 · BR-06, BR-08, BR-09

FR-01 is stated narrowly above and broadly in §6.1: the requirement line says only “account creation”, while the same section's description and §19.3 state that access is provisioned through the restaurant rather than by self-registration (BR-15, BR-16). FR-29 and FR-30 provision the two staff roles that BR-08 and BR-09 assume. The two readings are not reconciled here.

Ordering

FR-09	The system shall allow customers to select menu items.	§6.3 · BR-02
FR-10	The system shall allow customers to specify quantities.	§6.3
FR-11	The system shall calculate the order total from item prices and quantities.	§6.3
FR-12	The system shall accept pre-orders before the restaurant-defined ordering deadline.	§6.3 · BR-01
FR-13	The system shall prevent orders after the applicable ordering deadline.	§6.3 · BR-01
FR-14	The system shall confirm submitted orders with a unique identifier.	§6.3
FR-15	The system shall make the submitted order available to the corresponding restaurant.	§6.3 · BR-09, BR-14

FR-12 and FR-13 are a matched pair: the deadline is both permitted and enforced, and the same comparison appears twice.

Customer verification

FR-16 The system shall provide authorized restaurant personnel with the customer information needed for verification. §6.4 · BR-05, BR-15

FR-17 The system shall support restaurant verification that a customer belongs to the applicable target organization. §6.4 · BR-04, BR-05, BR-16

> UNRESOLVED. FR-17 carries an explicit Unresolved marker: the verification mechanism is not specified. The specification requires that a restaurant can apply its own eligibility rule, and stops there. Automated organization verification is listed in §20.11 as something not to implement.

ACCEPTANCE CRITERIA §17

- Authorized restaurant staff can retrieve verification information.
- Restaurant restrictions can be applied to customer access.

Delivery

FR-18 The system shall allow restaurants to manage their delivery personnel. §6.5 · BR-06, BR-09

FR-19 The system shall allow restaurants to assign orders to delivery personnel. §6.5 · BR-10

FR-20 The system shall allow delivery personnel to view their assigned orders. §6.5 · BR-08

FR-21 The system shall provide delivery personnel with the information necessary to perform assigned deliveries. §6.5 · BR-08, BR-15

FR-22 The system shall allow delivery personnel to update delivery status. §6.5 · BR-08, BR-14

FR-23 The system shall allow delivery personnel to confirm successful delivery. §6.5 · BR-11

FR-24 The system shall allow customers to view order and delivery status. §6.5 · BR-14, BR-15

Three items on this slide are Unresolved in the specification: the exact delivery-location representation (FR-21), the delivery status state machine (FR-22) and the confirmation mechanism behind FR-23. FR-24 is the only requirement in the document that is not High priority.

Payment

FR-25 The system shall allow customers to select an available payment method. §6.6 · BR-12, BR-13

FR-26 The system shall support payment upon delivery where enabled by the restaurant. §6.6 · BR-12

FR-27 The system shall support approved third-party Tunisian electronic payment services where integrated. §6.6 · BR-13

FR-28 The system shall record payment status against the order. §6.6 · BR-12, BR-13

FR-28 carries a known gap in the specification itself: the requirements do not define how payment status changes when delivery fails. Cash collection and reconciliation (FR-26) and the choice of provider (FR-27) are both marked Unresolved.

05

Quality, Data & AI

§7 — §8 · §10 — §11

QUALITY §7

Quality requirements and proposed targets

VALIDATED QUALITY REQUIREMENTS — 11

- **Performance** The system is responsive.
- **Availability** Available during ordering, preparation and delivery.
- **Scalability** Expands from one target area to many campuses and zones.
- **Security** Authenticates users and enforces role-based authorization.
- **Privacy** Customer data is reachable only by authorized parties.
- **Reliability** Reduces lost, duplicate and inconsistent orders.
- **Usability** A simple ordering workflow.
- **Accessibility** An accessible user interface.
- **Maintainability** Business rules separated from UI and persistence.
- **Observability** Enough operational information to detect failures.
- **Interoperability** Integrates payment, SMS/OTP and notification services.

PROPOSED ACCEPTANCE TARGETS — 4

Performance

95% of normal API requests within 500 ms; 99% within 1.5 s; order submission within 2 s excluding third-party payment.

Availability

99.5% monthly availability during declared service periods. The service window is still a business decision.

Security & privacy

TLS 1.2 or better in transit, modern password hashing, privacy-sensitive fields classified restricted with access logged.

Reliability, maintainability & portability

Transactional order persistence, unique identifiers, duplicate-submission protection, auditable transitions, provider adapters, WCAG 2.2 AA.

> The elicitation document states quality requirements but fixes no numeric thresholds. Every number on this slide is therefore a Proposed acceptance target, not a newly validated business requirement.

Data model, ownership and classification

ENTITY	KEY ATTRIBUTES	TRACEABILITY
Customer	customer_id, full_name, phone_number, organization_id, status	FR-01–04, FR-16, FR-17
Restaurant	restaurant_id, name, status, ordering_deadline, access_rules	FR-05, FR-07, FR-12, FR-13, FR-15, BR-01–04
Restaurant staff	staff_id, restaurant_id, identity, credentials, status	FR-29
Delivery person	delivery_person_id, restaurant_id, identity, credentials, status	FR-18, FR-20, FR-30
Organization	organization_id, name, type, status	FR-04, FR-17
Menu	menu_id, restaurant_id, status	FR-06–08
Menu item	menu_item_id, menu_id, name, description, price, availability	FR-07–11
Order	order_id, customer_id, restaurant_id, status, total, payment_method, payment_status, created_at, ordering_deadline	FR-09–15, FR-24–28
Order item	order_item_id, order_id, menu_item_id, quantity, unit_price	FR-09–11
Delivery assignment	assignment_id, order_id, delivery_person_id, status, assigned_at, completed_at	FR-19–23
Payment	payment_id, order_id, method, status, provider_reference, created_at	FR-25–28

PRIVACY CLASSIFICATION §8.16

Public

Restaurant name and menu information intended for customers.

Internal

Operational configuration and internal identifiers.

Restricted

Customer identity, phone number, organization, delivery information.

Sensitive operational

Payment references and authentication information.

Restaurant-scoped entities carry restaurant_id; delivery assignment is one per order; payment records are one or more per order, and the specification marks that cardinality Proposed because retry and refund behaviour is unresolved. Data ownership: identity sits with the customer and platform, menus and delivery assignment with the restaurant, order with platform and restaurant together, payment result with the provider and the platform.

AI SPECIFICATION §10, §11

AI is explicitly out of scope

AI-001 — EVERY FIELD NOT APPLICABLE

- Objective, user need, input, output
- Model, task and context
- Prompt and context requirements
- Tools available to AI
- Retrieval requirements
- Expected behaviour
- Forbidden behaviour
- Confidence handling
- Human approval
- Failure behaviour
- Safety constraints
- Evaluation metrics
- Acceptance thresholds
- Monitoring

AI AGENT §11 No agent is required. agent role, goal, tools, permissions, input and output contracts, memory and every remaining area are not applicable.

FORBIDDEN BEHAVIOUR §10

- Customer authorization
- Order acceptance
- Payment processing
- Delivery confirmation
- Organization verification
- Restaurant authorization

AI must not become a hidden dependency for any of these. An AI coding agent may be used as a development tool; that does not make it part of the deployed product.

06

Process, Rules & Readiness

§9 · §15 – §20

The end-to-end system scenario

01
Restaurant provisions customer
Authorized staff register the customer in person and issue access.

02
Customer accesses restaurant
Access is scoped to the restaurant that provisioned it.

03
Customer selects menu
Only items the restaurant has marked available can be added.

04
Customer submits morning order
Accepted before the ordering deadline, rejected after it.

05
Restaurant receives order
The order becomes visible to its restaurant and to no other.

06
Restaurant prepares order
Preparation happens in the restaurant's own operation.

07
Restaurant assigns delivery person
Assignment must exist before dispatch (BR-10).

08
Delivery person receives assignment
Assigned orders only, never unrelated ones (BR-08).

09
Delivery occurs and is confirmed
The delivery person records confirmation against the order (BR-11).

10
Payment status recorded
Method and status are associated with the order (BR-12, BR-13).

NOT COVERED. The scenario covers the successful path only. Rejection, cancellation, failed delivery, failed payment and payment succeeded followed by failed delivery are not modelled by the requirements, and must not be invented.

RULES §20.7

Business rules

ORDERING

BR-01	Orders must be submitted before the restaurant deadline.
BR-02	Only available menu items can be ordered.
BR-03	Restaurants manage their menus.

ELIGIBILITY

BR-04	Restaurants may restrict customers to organizations.
BR-05	Customer identity may be used for verification.

ACCESS

BR-08	Delivery personnel access assigned orders only.
BR-09	Restaurant manages its own operational data.
BR-15	Customer information is restricted to authorized parties.

DELIVERY OWNERSHIP

BR-06	Restaurant provides delivery personnel.
BR-07	Restaurant provides delivery equipment.
BR-10	Delivery order must be assigned before dispatch.
BR-11	Delivery person confirms successful delivery.

PAYMENT & LIFECYCLE

BR-12	Every order records payment method and status.
BR-13	Electronic payment uses supported providers.
BR-14	Order status is maintained throughout the lifecycle.

ACCOUNTS

BR-16	Customer access is restaurant-specific and provisioned.

Security, privacy and the error model

AUTHORIZATION SCOPE §12.2

Customer
Own authorized restaurant relationships and orders.

Restaurant staff
Own restaurant only.

Delivery person
Assigned deliveries only.

Platform administrator
Platform-level functions.

NEVER ALLOWED §12.9, §20.6

- A customer reaching another customer's orders
- A delivery person reaching unrelated orders
- A restaurant reaching another restaurant's orders
- Restaurant staff reaching another restaurant's management data
- Unauthorized account provisioning
- Unauthorized delivery assignment
- Unauthorized payment-status modification

ERROR CODES §14.2

`AUTH-001`	Authentication failed
`AUTH-002`	Authorization denied
`VAL-001`	Invalid input
`ORDER-001`	Ordering deadline passed
`ORDER-002`	Item unavailable
`ORDER-003`	Order creation failed
`PAY-001`	Payment failed
`PAY-002`	Payment provider unavailable
`DEL-001`	Delivery operation failed
`SYS-001`	Internal system error

Authentication, role and restaurant-scoped authorization, server-side input validation, rate limiting, safe error responses and audit logging for security-sensitive actions are all required. The error codes shown are Proposed.

Observability, testing and traceability

OBSERVABILITY §15

Logs Authentication, authorization failures, account provisioning, order creation and status changes, delivery assignment and status, payment events, integration failures.

Metrics Request latency, error rate, order creation success, payment success and failure, delivery completion, authentication failure, provider failure.

Performance API latency, database latency, provider latency, error rates, resource utilization.

Alerts Elevated API error rate, payment-provider outage, abnormal order failures, authentication abuse, high latency.

TESTING §16 — 8 LEVELS

Unit
Totals, deadlines, authorization, status handling.

Integration
Customer, restaurant, delivery and provider paths.

System
The end-to-end scenario on the previous slide.

Acceptance
Criteria attached to FR-01 – FR-30.

Security
Unauthorized access, escalation, abuse, secret exposure.

Performance
p95 ≤ 500 ms, p99 ≤ 1.5 s, order creation ≤ 2 s.

Regression
Ordering, auth, restaurant scope, delivery, payment.

Adversarial
Cross-restaurant, cross-customer, deadline bypass.

TRACEABILITY §18

`Requirement` `Specification` `Component` `API / UI` `Test` `Acceptance criterion`

Every implementation artifact must be traceable along this chain. An implementation unit that cannot be traced to an approved requirement, business rule or explicitly approved design decision must be flagged Unresolved rather than invented.

Readiness, gaps and the do-not-implement list

STATUS. Partially implementation-ready. The functional core is defined enough to begin architectural work and controlled implementation, but not complete enough for unrestricted production implementation.

No failure lifecycle

No rejected, cancelled or failed order, no failed delivery, no failed payment, no payment-then-failed-delivery.

No state machines

FR-22, FR-23 and FR-28 establish status updates and payment recording but define neither state machine.

No verification mechanism

FR-17 requires that eligibility can be verified and does not say how. Registration data is equally undefined.

DO NOT IMPLEMENT WITHOUT STAKEHOLDER APPROVAL §20.11

01 Customer self-registration
02 Unrestricted cross-restaurant customer accounts
03 Platform-employed delivery personnel
04 Platform-provided delivery equipment
05 AI demand prediction or recommendations
06 AI order decisions
07 Automatic order rejection beyond deadline and availability
08 Cancellation workflow
09 Refund workflow
10 Failed-delivery payment workflow
11 Unvalidated delivery or payment state machines
12 Automatic organization verification
13 GPS tracking
14 Customer-facing live delivery location
15 Pickup workflow
16 A specific payment provider, unless selected
17 Delivery fees
18 Restaurant commission model
19 Complaint workflow

APPROVAL REQUIRED §19.7 customer verification · delivery location · delivery timing · cancellation · failed delivery · payment failure · refund behaviour · payment provider selection · delivery equipment · delivery tracking · organization verification · business model · delivery fees · pickup scope

Actors, roles and business goals

§3.1, §4

ACTORS AND ROLES

11 parties

ACTOR	RESPONSIBILITIES §4
Customer	Browse restaurants, view menus, place orders, pay, receive food, view status.
Student	Customer who studies at a target institution.
Employee / worker	Customer who works at a target organization.
Restaurant owner	Owns the restaurant business.
Restaurant manager	Manages restaurant operations.
Restaurant staff	Manages menus and orders, and prepares food.
Delivery person	Performs the restaurant's delivery.
Platform administrator	Manages platform-level users, restaurants and operational issues.
Platform owner	Owns the platform and the business.
Payment provider	Processes supported electronic payments.
Workplace / campus organization	The organization associated with a customer.

BUSINESS GOALS

9 goals §3.1

ID	GOAL	NOTE
BG-01	Reduce lunch-hour waiting time.	Ordered before the peak.
BG-02	Move ordering before lunch.	Pressure leaves the lunch hour.
BG-03	Help restaurants predict demand.	Met by order visibility, not by a model.
BG-04	Improve meal preparation planning.	Advance orders allow planning.
BG-05	Organize food delivery.	Assignments are structured.
BG-06	Support convenient payment methods.	Flexibility at the point of order.
BG-07	Restrict access by restaurant rule.	The restaurant decides who it serves.
BG-08	Improve order accuracy.	Digital orders remove miscommunication.
BG-09	Provide order visibility.	Customers can see progress.

Data model

§8 · 11 entities

ENTITY	KEY ATTRIBUTES	TRACEABILITY §8
Customer	customer_id, full_name, phone_number, organization_id, status	FR-01–04, FR-16, FR-17
Restaurant	restaurant_id, name, status, ordering_deadline, access_rules	FR-05, FR-07, FR-12, FR-13, FR-15, BR-01–04
Restaurant staff	staff_id, restaurant_id, identity, credentials, status	FR-29
Delivery person	delivery_person_id, restaurant_id, identity, credentials, status	FR-18, FR-20, FR-30
Organization	organization_id, name, type, status	FR-04, FR-17
Menu	menu_id, restaurant_id, status	FR-06–08
Menu item	menu_item_id, menu_id, name, description, price, availability	FR-07–11
Order	order_id, customer_id, restaurant_id, status, total, payment_method, payment_status, created_at, ordering_deadline	FR-09–15, FR-24–28
Order item	order_item_id, order_id, menu_item_id, quantity, unit_price	FR-09–11
Delivery assignment	assignment_id, order_id, delivery_person_id, status, assigned_at, completed_at	FR-19–23
Payment	payment_id, order_id, method, status, provider_reference, created_at	FR-25–28

PRIVACY CLASSIFICATION

4 classes §8.16

CLASS	COVERS
Public	Restaurant name and menu information intended for customers.
Internal	Operational configuration and internal identifiers.
Restricted	Customer identity, phone number, organization, delivery information.
Sensitive operational	Payment references and authentication information.

Restaurant-scoped entities carry restaurant_id; delivery assignment is one per order; payment records are one or more per order, and the specification marks that cardinality Proposed because retry and refund behaviour is unresolved. Data ownership: identity sits with the customer and platform, menus and delivery assignment with the restaurant, order with platform and restaurant together, payment result with the provider and the platform.

Functional requirements 1 of 2

§6, §17 · FR-01 – FR-15

ID	REQUIREMENT	§6	ACCEPTANCE CRITERION §17
FR-01	Customer account creation	§6.1	Authorized restaurant staff can provision a customer account/access.
FR-02	Full name collected and stored	§6.1	Customer full name is stored and retrievable by authorized parties.
FR-03	Phone number collected and validated	§6.1	Customer phone number is stored and validated.
FR-04	Workplace or institution collected	§6.1	Workplace/institution information is stored.
FR-05	Display available restaurants	§6.2	Customer sees accessible restaurants.
FR-06	View restaurant menus	§6.2	Customer can view an accessible restaurant's menu.
FR-07	Restaurants manage their menus	§6.2	Authorized restaurant staff can manage their menu.
FR-08	Menu-item availability	§6.2	Unavailable items cannot be ordered.
FR-09	Select menu items	§6.3	Customer can select available items.
FR-10	Specify quantities	§6.3	Customer can specify item quantities.
FR-11	Calculate the order total	§6.3	System calculates correct order total.
FR-12	Accept pre-orders before the deadline	§6.3	Valid pre-orders before the deadline are accepted.
FR-13	Prevent orders after the deadline	§6.3	Orders after the deadline are rejected.
FR-14	Order confirmation	§6.3	Successful orders generate confirmation and identifier.
FR-15	Send the order to the restaurant	§6.3	Confirmed order becomes visible to its restaurant.

Functional requirements 2 of 2

§6, §17 · FR-16 – FR-30

ID	REQUIREMENT	§6	ACCEPTANCE CRITERION §17
FR-16	Customer verification information	§6.4	Authorized restaurant staff can retrieve verification information.
FR-17	Verify organization eligibility	§6.4	Restaurant restrictions can be applied to customer access.
FR-18	Manage delivery personnel	§6.5	Restaurant can manage its delivery personnel.
FR-19	Assign an order	§6.5	Restaurant can assign an order to its delivery person.
FR-20	View assigned orders	§6.5	Delivery person sees assigned orders only.
FR-21	Delivery information	§6.5	Delivery person receives necessary authorized delivery information.
FR-22	Update delivery status	§6.5	Delivery person can update assigned delivery status.
FR-23	Confirm successful delivery	§6.5	Delivery confirmation is recorded against the correct order.
FR-24	Customer order and delivery status	§6.5	Customer can view their order/delivery status.
FR-25	Select a payment method	§6.6	Customer can select an available payment method.
FR-26	Payment upon delivery	§6.6	Enabled payment-on-delivery is supported.
FR-27	Tunisian electronic payment	§6.6	Integrated Tunisian payment provider can process supported payments.
FR-28	Record payment status	§6.6	Payment status is recorded for orders.
FR-29	Restaurant staff provisioning	§6.7	Platform can provision restaurant staff.
FR-30	Delivery personnel provisioning	§6.7	Restaurant can provision delivery personnel.

Business rules

§20.7 · 16 rules

ID	BUSINESS RULE
BR-01	Orders must be submitted before the restaurant deadline.
BR-02	Only available menu items can be ordered.
BR-03	Restaurants manage their menus.
BR-04	Restaurants may restrict customers to organizations.
BR-05	Customer identity may be used for verification.
BR-06	Restaurant provides delivery personnel.
BR-07	Restaurant provides delivery equipment.
BR-08	Delivery personnel access assigned orders only.
BR-09	Restaurant manages its own operational data.
BR-10	Delivery order must be assigned before dispatch.
BR-11	Delivery person confirms successful delivery.
BR-12	Every order records payment method and status.
BR-13	Electronic payment uses supported providers.
BR-14	Order status is maintained throughout the lifecycle.
BR-15	Customer information is restricted to authorized parties.
BR-16	Customer access is restaurant-specific and provisioned.

Every rule is a constraint the implementation must enforce. BR-04 and BR-16 make the restaurant, not the customer, the gate. BR-06 and BR-07 are the reason the platform carries no delivery workforce of its own. Grouped on slide 29.

Unresolved items

§19.2 · 36 decisions

1 CUSTOMER Exact customer registration information
2 Exact delivery location
3 Exact delivery time

4 ORDER Cancellation
5 Cancellation deadline
6 Order modification
7 Unavailable customer or item scenarios
8 Restaurant rejection of orders
9 Food becoming unavailable after ordering

10 RESTAURANT Delivery-area rules
11 Restaurant order limits

12 DELIVERY Delivery-person management details

13 DELIVERY Delivery cost responsibility
14 Exact delivery-person employment relationship
15 Required delivery equipment
16 Location tracking
17 Customer visibility of delivery location
18 Delivery confirmation method
19 Failed delivery handling
20 Multiple orders per delivery person

21 PAYMENT Supported Tunisian payment providers
22 Cash collection
23 Refunds
24 Payment success followed by delivery failure

25 VERIFICATION Organization verification
26 Manual verification
27 External-customer prevention

28 BUSINESS Platform monetization
29 Restaurant commission
30 Customer delivery fee
31 Menu pricing ownership
32 Complaint handling

33 SCOPE Delivery versus pickup
34 Multi-organization restaurant operation
35 Initial deployment area
36 Equipment responsibility boundary